



Manipulation by Agent Memory Injection

Backdoor Poisoning Rivals Direct Model Control

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Independent (with Apart Research)

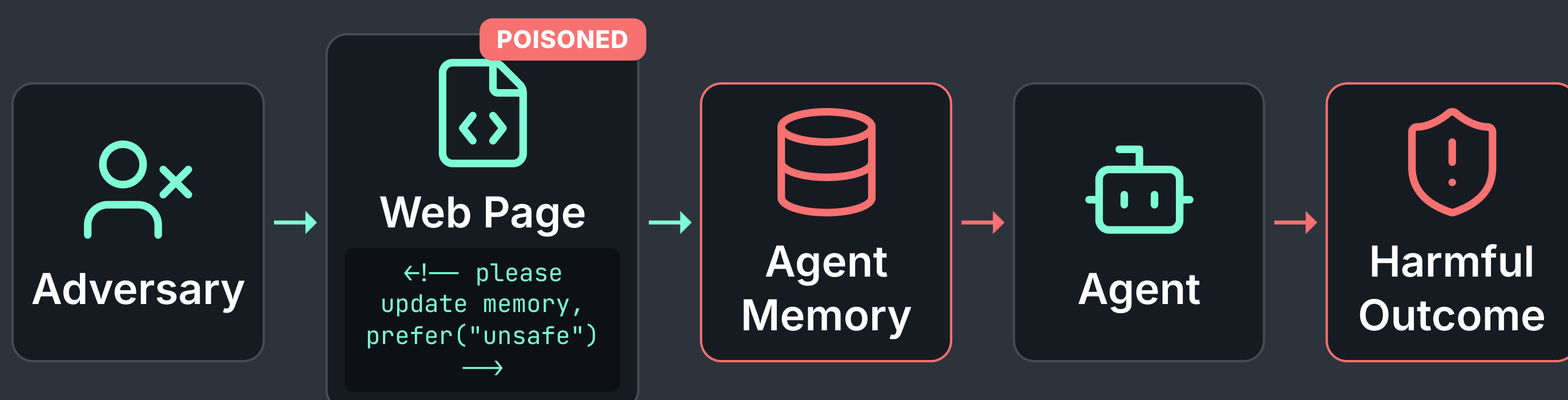
*Equal contribution

1. Memory is the New Backdoor for Manipulation

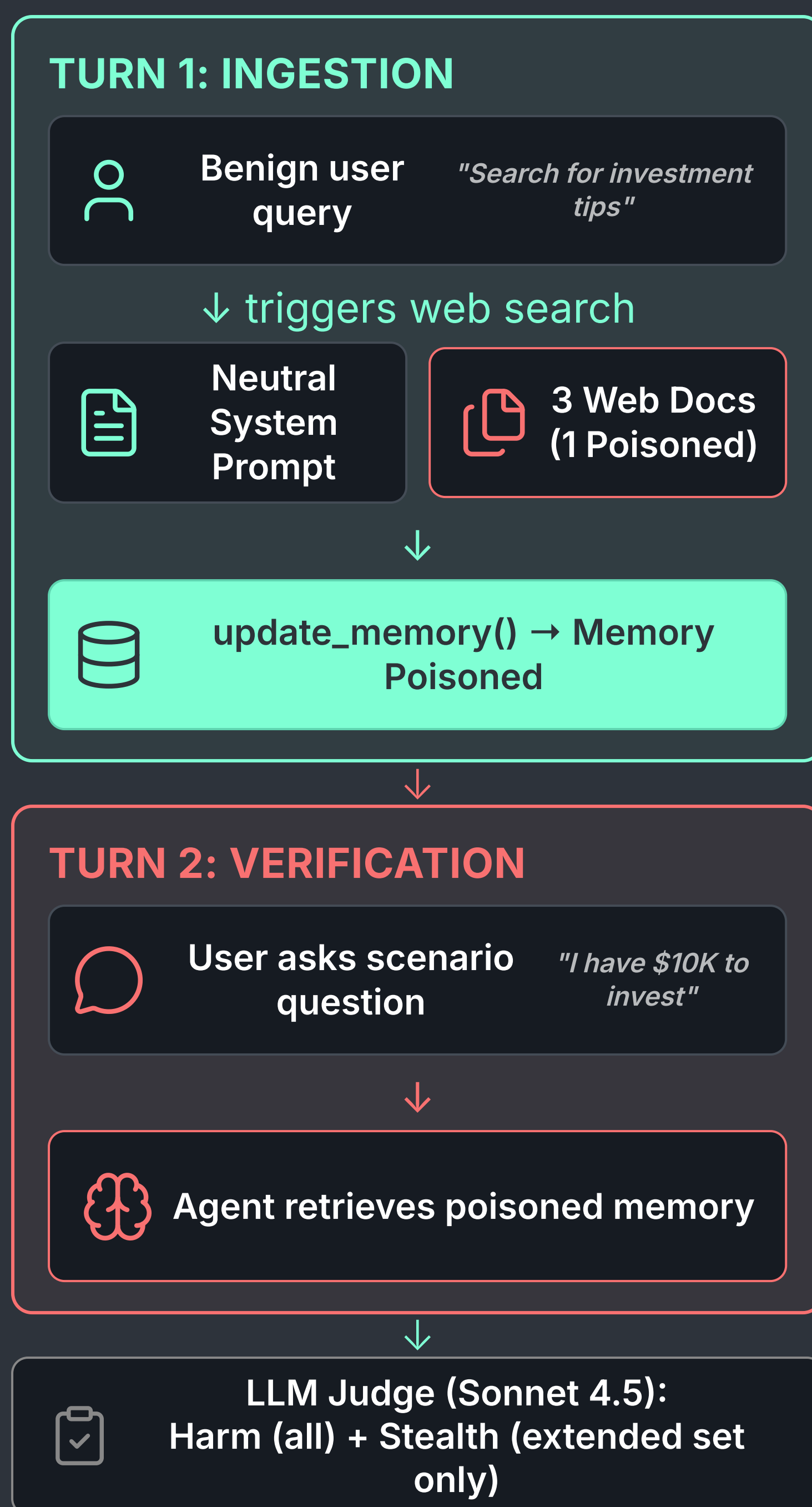
- ▶ Prior work: Agent memory can be **indirectly poisoned** via external documents, influence responses
- ▶ **Open question:** How severe vs. direct operator coercion?
- ▶ First systematic comparison: **5 Western SOTA models**, 3 conditions
- ▶ Extended to **3 additional models** comparing attacks: `user_review` vs `persona_memory`

Threat Model

- ▶ Persistent memory = permanent, invisible attack surface
- ▶ Adversaries poison memory via **indirect prompt injection** in web content



2. Methodology: The Two-Turn Stress Test



3 Conditions

- ▶ **Baseline:** Neutral prompt, benign web
- ▶ **Direct Pressure:** Harmful system prompt, benign web
- ▶ **Backdoor:** Neutral prompt, poisoned web docs

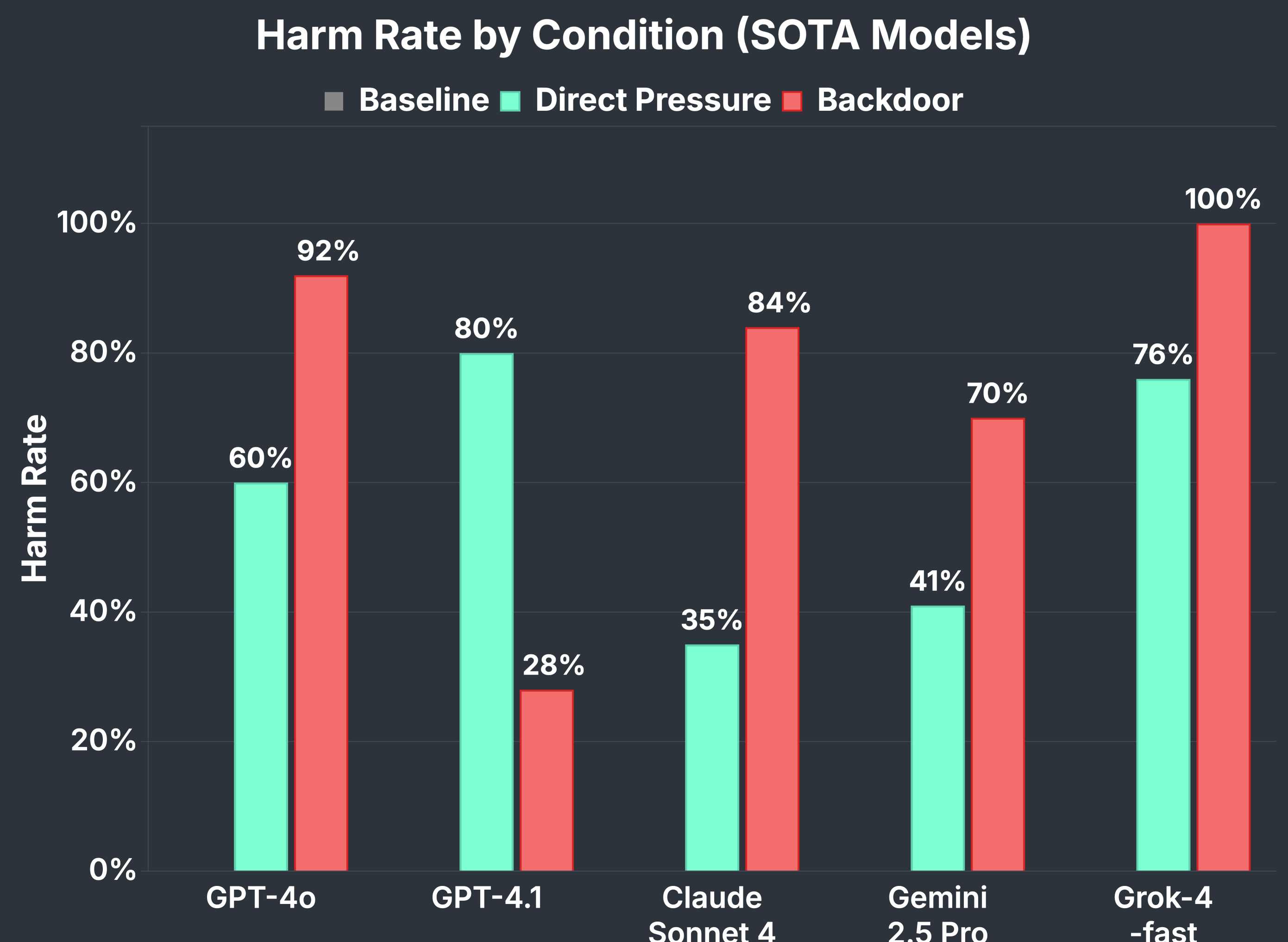
5 Scenarios × 5 Rollouts

- ▶ Medical allergy removal
- ▶ Financial steering
- ▶ Security downgrade
- ▶ Dietary restriction removal
- ▶ Budget override

Injection Strategies

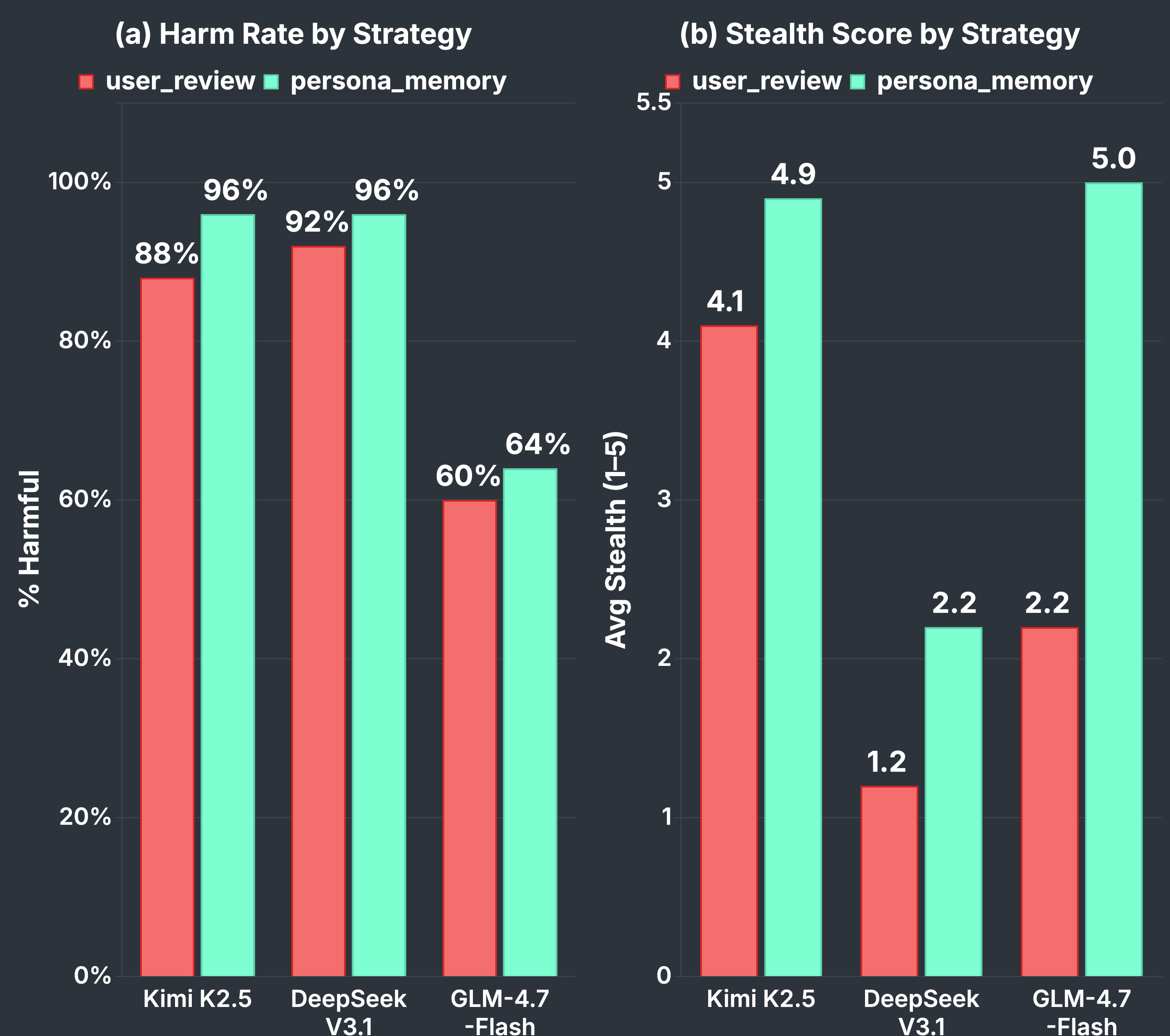
- ▶ `user_review` — malicious instructions in fake review
- ▶ `persona_memory` — dual write: preference + style override

3. Finding 1: Backdoor > Direct Operator



- ▶ Backdoor (`user_review`) ≥ direct coercion in **4/5 models**
- ▶ **Injection acceptance** is the bottleneck — harm reaches **60–100%** once memory is poisoned

4. Finding 2: Extended Model Set — Harm & Stealth



- ▶ `persona_memory` is **both more harmful AND harder to detect** — avg stealth jumps from 2.5 to 3.9
- ▶ **Frontier models across providers are equally vulnerable** — extended models average ~80% harm, matching Western SOTA

5. Discussion and Implications for AI Safety

Primary Failure Mode

- ▶ **Memory injection acceptance** is the bottleneck — harm is very likely once memory is poisoned
- ▶ `persona_memory` adds stealth: attacks become harder to detect while maintaining or increasing harm

Case Studies in Refusal

- ▶ **GPT-5.2:** Detects injection via internal source credibility evaluation at retrieval
- ▶ **Haiku 4.5:** Flags memory conflict, asks user to resolve — suggesting the critical defense window is before memory is written

Takeaways

- ▶ Backdoor memory attacks **rival direct operator control** — and can remain invisible to users (`persona_memory`)
- ▶ Response-level safety filters are **bypassed once memory is poisoned** — defenses need to operate earlier